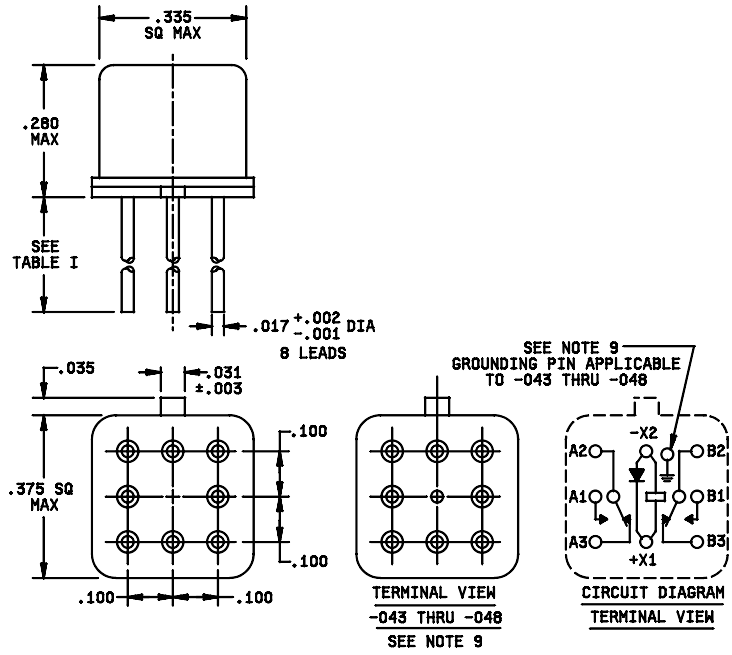
INCH-POUND
MIL-PRF-39016/19H
1 April 2005
SUPERSEDING
MIL-PRF-39016/19G
20 July 1988

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT, LOW LEVEL TO 1 AMPERE WITH INTERNAL DIODE FOR COIL TRANSIENT SUPPRESSION AND POLARITY REVERSAL PROTECTION, TERMINALS 0.100-INCH GRID PATTERN

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the relays described herein shall consist of this specification sheet and the latest issue of MIL-PRF-39016.



Inches	mm
.001	0.03
.002	0.05
.003	0.08
.017	0.43
.031	0.79
.035	0.89
.100	2.54
.280	7.11
.335	8.51
.370	9.40

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above are for reference only. Numbers do not appear on relay.
5. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
6. Coil symbol optional in accordance with MIL-STD-1285.
7. Circuit diagram shown on part is the terminal view.
8. See table I for lead lengths.
9. M39016/19-043 through M39016/19-048 shall be supplied with a case grounding pin welded to the relay header as shown.

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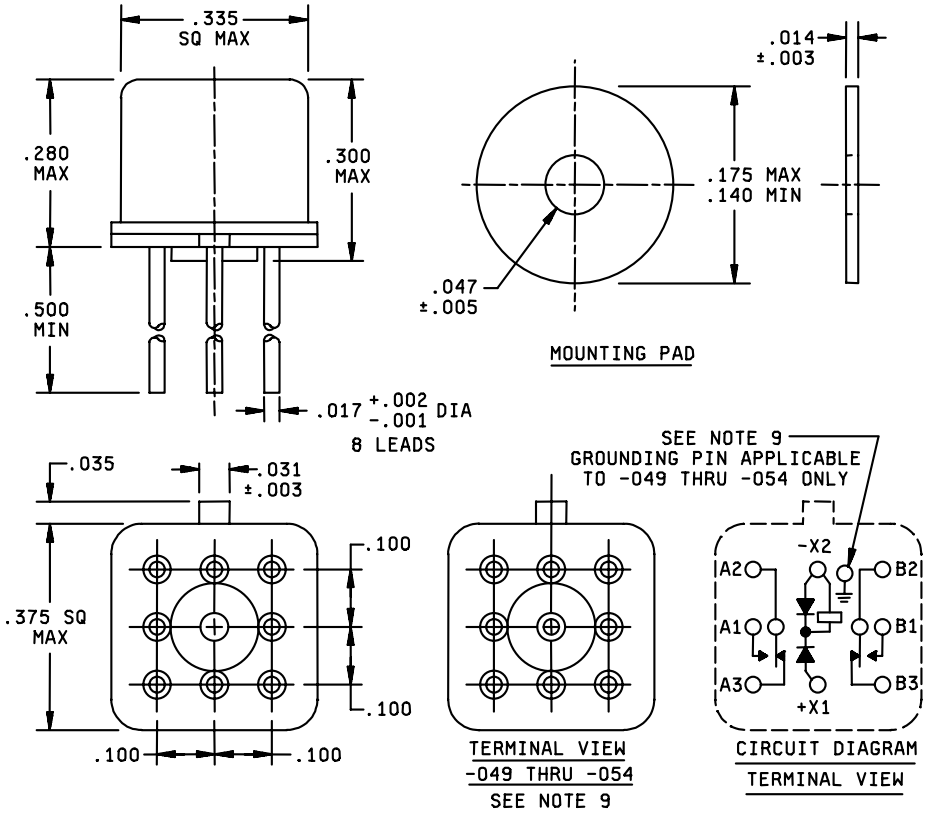


FIGURE 1. Dimensions and configuration.

Inches	mm
.001	0.03
.002	0.05
.003	0.08
.005	0.13
.014	0.36
.017	0.43
.031	0.79
.035	0.89
.047	1.19
.100	2.54
.170	4.32
.280	7.11
.300	7.62
.335	8.51
.370	9.40
.500	12.70

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above for reference only. Numbers do not appear on relay.
5. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
6. Coil symbol optional in accordance with MIL-STD-1285.
7. Circuit diagram shown on part is the terminal view.
8. Spacer mounting pad shall be a polyester film per MIL-I-631, Type G, Class I.
9. M39016/19-049 through -054 shall be supplied with a case grounding pin welded to the relay header as shown.

FIGURE 2. Dimensions and configuration (relay with spacer mounting pad).

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REQUIREMENTS:

CONTACT DATA:

Load ratings:

High level (relay case grounded):

Resistive:
1.0 ampere at 28 V dc.
250 milliamperes at 115 V ac 60 and 400 Hz case not grounded.
100 milliamperes at 115 V ac 60 and 400 Hz case grounded.

Inductive load: 0.2 ampere at 28 V dc with 0.32 henry inductance.

Lamp: 0.10 ampere at 28 V dc.

Low level: 10 to 50 μA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance or voltage drop:

Initial: 0.10 ohm maximum (0.110 ohm maximum with spacer mounting pad).

High level:

During life: Not more than 5 percent of open circuit voltage.

After life: 0.20 ohm maximum (0.210 ohm maximum with spacer mounting pad).

Low level:

During life: 100 ohms maximum.

After life: 0.15 ohm maximum (0.160 ohm maximum with spacer mounting pad).

Intermediate current:

During: 3 ohms maximum.

After: 0.20 ohm maximum (0.210 ohm maximum with spacer mounting pad).

Contact bounce: 1.5 milliseconds maximum (applicable to failure rate level “L”).

Contact stabilization time: 2.5 milliseconds maximum (applicable to failure rate levels “M”, “P”, and “R”).

Overload (high level only): Two times rated current. Not applicable to ac load ratings.

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COIL DATA: See table I.

Operate time: 2.0 ms maximum over temperature range with rated coil voltage.

Release time: 4.0 ms maximum over temperature range from rated coil voltage.

ELECTRICAL DATA:

Insulation resistance: 10,000 megohms minimum at 500 V dc, except the resistance between coil and case at high temperature shall be 1,000 megohms minimum.

Dielectric withstanding voltage:

	Sea level V rms (60 Hz)	Post intermediate current life test Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure and all contacts in the energized and de-energized positions.	500	500	
Between case, frame, or enclosure and coil.	500	500	125
Between all contacts and coils.	500	500	All terminals to case
Between open contacts in the energized and de-energized positions.	500	375	
Between contact poles in the energized and de-energized positions.	500	500	
Between coils of dual coil relays.	N/A	N/A	

DIODE CHARACTERISTICS

Coil transient suppression: Applicable.

Diode block integrity (perform this test after coil transient suppression test in all inspection tables of MIL-PRF-39016): With applicable voltage applied to the relay coil circuit in the reverse direction, monitor leakage current with dc microammeter, oscilloscope, or qualifying activity approved test equipment. Leakage current shall not exceed the specified value.

Block integrity maximum leakage current: 1μA at 50 V dc.

Maximum negative transient: 1.0 volt.

Breakdown voltage: 100 V dc minimum at 10 microamperes (μA). (This test may be performed in-process or as final assembly).

Semiconductor in-process screening: Applicable, visual inspection of semiconductors shall be in accordance with MIL-STD-750, method 2073 or 2074.

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Shock (specified pulse): MIL-STD-202, method 213, test condition B (75 g's). Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

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Vibration (sinusoidal): MIL-STD-202, method 204. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

Vibration (random): MIL-STD-202, method 214, test condition IG. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts. Applicable to qualification and group C testing only.

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Salt atmosphere (corrosion): In accordance with method 1041, MIL-STD-750.

PHYSICAL DATA:

Terminal strength (MIL-STD-202, method 211):

Pull test: Test condition A, 1 pound pull.

Bend test: Test condition C, ½ pound load.

Twist test: As specified in MIL-PRF-39016.

Solderability: Applicable.

Dimensions and configuration: See figure 1.

Weight: 2.25 grams (0.09 ounce) maximum.

Seal: Hermetic.

Minimum marking: Military part number, “J” with the date code (example J0430), circuit diagram, manufacturer’s name or source code.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles per relay.

Low level: 100,000 cycles plus 900,000 cycles mechanical life.

Part or Identifying Number (PIN): M39016/19- (dash number from table I and suffix letter designating failure rate level).

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TABLE I. Dash numbers and characteristics. 1/

Dash number 2/			Coil voltages (V dc)		AT +25°C						Over temperature range		
Lead length .187 ±.010	Lead length .500 min	Spacer mounting pads (fig. 2) 3/	Rated	Max	Coil resistance (ref. only) Ohms 4/	Coil circuit current (mA) 5/ 7/ Max	Coil circuit current (mA) 5/ 7/ Min	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified dropout value (voltage) (V dc)	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified dropout value (voltage) (V dc)
025	031	037	5.0	5.8	39	128.2	93.2	3.2	2.3	0.6	4.0	2.8	0.6
026	032	038	6.0	8.0	78	78.3	58.3	4.0	2.8	0.7	5.0	3.4	0.7
027	033	039	9.0	12.0	220	42.9	33.0	6.3	4.2	0.9	7.8	5.3	0.8
028	034	040	12.0	16.0	390	32.8	25.6	8.0	5.2	1.1	10.0	6.5	0.9
029	035	041	18.0	24.0	880	22.1	17.5	11.5	7.3	1.4	14.5	10.0	1.1
030	036	042	26.5	32.0	1,560	18.5	14.8	15.2	9.5	1.8	19.0	13.0	1.4
	043 6/	049 6/	5.0	5.8	39	128.2	93.2	3.2	2.3	0.6	4.0	2.8	0.6
	044 6/	050 6/	6.0	8.0	78	78.3	58.3	4.0	2.8	0.7	5.0	3.4	0.7
	045 6/	051 6/	9.0	12.0	220	42.9	33.0	6.3	4.2	0.9	7.8	5.3	0.8
	046 6/	052 5/	12.0	16.0	390	32.8	25.6	8.0	5.2	1.1	10.0	6.5	0.9
	047 6/	053 6/	18.0	24.0	880	22.1	17.5	11.5	7.3	1.4	14.5	10.0	1.1
	048 6/	054 6/	26.5	32.0	1,560	18.5	14.8	15.2	9.5	1.8	19.0	13.0	1.4

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits not recommended for subsequent use in low level applications.

2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): L, 3.0; M, 1.0; P, 0.1; R, 0.01. Example, 031L - - - - - - -42R.

3/ Relays supplied with spacer mounting pads (-037 through -042 and -049 through -054) shall have the spacer mounting pad rigidly attached.

4/ CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.

5/ Coil resistance not directly measurable at relay terminals. When rated voltage is applied to the coil terminals, the coil circuit current must be within the limits shown. Measure at 25°C at nominal voltage for 5 seconds, maximum.

6/ Relays are supplied with a case ground pin welded to the header (see figures 1 and 2).

7/ Delete “Coil resistance” and substitute “Coil current” test in all inspection tables of MIL-PRF-39016.

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QUALIFICATION INSPECTION:

Qualification inspection and sample size: See table II.

TABLE II. Qualification inspection and sample size. 1/

Single submission	Group submission	
18 units plus 1 open unit for level L at C = 0 2/	M39016/19-036	18 units plus 1 open unit for level L at C = 0 2/
33 units plus 1 open unit for level M at C = 0 2/		33 units plus 1 open unit for level M at C = 0 2/
Qualification inspection as applicable		Qualification inspection as applicable
	M39016/19-031 M39016/19-032 M39016/19-033 M39016/19-034 M39016/19-035	2 units each PIN qualification inspection, Q2.
	M39016/19-048	1 unit terminal strength and solderability.

1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-PRF-39016/43 may be used in addition to MIL-PRF-39016/19 data. Prior to performance of retention of qualification testing, the relay manufacturer shall preselect the sampling plan.

2/ The number of units required for qualification testing shall be increased as required in Q5, MIL-PRF-39016, if the relay manufacturer elects to test the number of units permitting one or more failures. Prior to performance of qualification testing, the relay manufacturer shall preselect the sampling plan.

Qualification inspection (reduced testing) (sample size - 2 size each coil voltage and 1 unsealed unit). See table III.

If the relays produced for MIL-PRF-39016/19 are similar in construction and design except for the coils to the relays produced for MIL-PRF-39016/43, then reduced testing for qualification of MIL-PRF-39016/19 relays may be performed concurrent with or subsequent to successful qualification of MIL-PRF-39016/43 relays. For reduced testing, see table III.

TABLE III. Qualification inspection (reduced testing).

Examination or test
2 units each coil voltage Q2 of qualification inspection table 1 unsealed sample unit for internal examination.

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Initial qualification of relays supplied with spacer mounting pads (-037 through -042 and -049 through -054), shall be tested as specified below:

Perform the following tests as specified in the qualification inspection table of MIL-PRF-39016, in the order shown below:

Before installation of spacer mounting pad: Screening, visual and mechanical examination (internal), thermal shock, resistance to solvents, vibration (sinusoidal), vibration (random), shock (specified pulse), acceleration, terminal strength, magnetic interference (when specified), capacitance (when specified), coil life (applicable to continuous duty relays only), resistance to soldering heat, salt spray (corrosion), overload (applicable to high level relays only), life, terminal strength, and intermediate current.

After installation of spacer mounting pad, perform the following tests as specified in the qualification inspection table of MIL-PRF-39016, in the order shown below:

Insulation resistance, dielectric withstanding voltage, static contact resistance, specified pickup, hold and dropout (value voltages), coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression (when specified), solderability, seal, visual and mechanical inspection (external).

Qualification inspection (reduced testing for previously qualified relays) for relays supplied with spacer mounting pads (-037 through -042 and -049 through -054): Two units of the 26.5-volt rated coil voltage (-042) shall be tested as specified below:

Before installation of spacer mounting pad, perform the following tests as specified in the qualification inspection table of MIL-PRF-39016 in the order shown below:

For failure rate level L only: Screening.

For failure rate levels M, P, and R: Vibration (sinusoidal) test duration shall be 10 minutes, vibration (random), screening.

After installation of spacer mounting pad, perform the following tests as specified in the qualification inspection table of MIL-PRF-39016 in the order shown below:

Insulation resistance, dielectric withstanding voltage, static contact resistance, specified pickup, hold, and dropout values (voltages), coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression (when specified), solderability, seal, visual and mechanical inspection (external).

Group A testing for relays supplied with spacer mounting pads (-037 through -042) shall be tested as specified below:

Perform seal test immediately preceding the A2 electrical tests. Relay leads shall be formed and the mounting pad removed before the seal test. After the seal test, the mounting pad shall be rigidly attached to the relay and the remaining group A tests performed (The seal test is not performed with group A4.

SUPERSESSION DATA:

Supersession data: See table IV.

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TABLE IV. Supersession data. 1/

Superseded part no. M39016/19-	New part no. M39016/19-		Superseded part no. M39016/19-	New part no. M39016/19-
001	031		013	031
002	032		014	032
003	033		015	033
004	034		016	034
005	035		017	035
006	036		018	036
007	025		019	025
008	026		020	026
009	027		021	027
010	028		022	028
011	029		023	029
012	030		024	030

1/ Dash numbers -001 through -024, .225 inch high cans have been canceled and are superseded by dash numbers -025 through -036, .280 inch high cans. The .225 inch cans are no longer manufactured.

Cross-reference for Government logistical support: See table V.

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TABLE V. Cross reference for Government logistical support.

Superseded part no. M39016/19-	New part no. M39016/19-	Support with part no. M39016/19-		Superseded part no. M39016/19-	New part no. M39016/19-	Support with part no. M39016/19-		New part no. M39016/19-	Support with part no. M39016/19-
001	031	031		022	028	034		043	043
002	032	032		023	029	035		044	044
003	033	033		024	030	036		045	045
004	034	034			025	031		046	046
005	035	035			026	032		047	047
006	036	036			027	033		048	048
007	025	031			028	034		049	049
008	026	032			029	035		050	050
009	027	033			030	036		051	051
010	028	034			031	031		052	052
011	029	035			032	032		053	053
012	030	036			033	033		054	054
013	031	031			034	034			
014	032	032			035	035			
015	033	033			036	036			
016	034	034			037	037			
017	035	035			038	038			
018	036	036			039	039			
019	025	031			040	040			
020	026	032			041	041			
021	027	033			042	042			

Referenced documents. In addition to MIL-PRF-39016, this document references the following:

MIL-I-631
MIL-PRF-39016/43
MIL-STD-202
MIL-STD-750
MIL-STD-1285

Changes from previous issue: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Army - AR
Navy - AS, MC, OS, SH
Air Force - 99

(Project 5945-1285)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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